INCH - POUND

MIL-I-50857A(MI)
27 June 1991

SUPERSEDING
MIL-I-50857(MI)
30 April 1971

MILITARY SPECIFICATION
INNER TUBE ASSEMBLY AND CONVERSION KIT, PARTS FOR

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers special requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembly and packing of one inner tube assembly and conversion kit for use with rocket, practice 35mm sub-caliber, M73.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Ammunition Lot Numbering
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-45662	-	Calibration Systems Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

9256067	-	Inner Tube Assembly and Conversion Kit
9256076	-	Box, Fiberboard, Packing, Ammunition for Inner Tube Assembly and Conversion Kit
9256075	-	Box, Packing, Ammunition for Inner Tube Assembly and Conversion Kit

PUBLICATIONS

ET 9256067	-	Equipment Tabulation

(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

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3.2 Parts. The parts shall comply with all requirements specified on Drawing (dwg.) 9256067 and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Inner tube assembly. The front and rear supports must withstand an axial pull-off force of 50+2 pounds after swaging.

3.5 Workmanship. All parts, subassemblies and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter that may impair proper functioning of the launcher or that may cause premature functioning of the loaded rocket. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used. Painted or coated surfaces shall be restored when these surfaces have been marred or scratched. All required markings shall be neat, clean, and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

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b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4 and 4.5.

4.4 First article inspection.

4.4.1 Submission. Prior to initiating regular production, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer.

4.4.2 Inspections to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.5.2 and 4.5.3 of this specification and any or all requirements of the applicable drawing.

4.4.3 Rejection. If the assembly or any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon observing any non-conformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.5 Quality conformance inspections.

4.5.1 Lot formation. The term "lot" as used throughout this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and specification revisions. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of

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an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.5.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.4.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for major defects shall be 100 percent. Sampling plans for major defects shall be in accordance with tables I, II, and III.

TABLE I. Sampling plan A.

Lot Size	Sample Size
2 to 32	All
33 to 280	32
281 to 500	48
501 to 3200	73
3201 to 10,000	86
10,001 to 35,000	108
35,001 to 150,000	123
150,001 to 500,000	156
500,001 and over	189
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE II. Sampling plan B.

Lot Size	Sample Size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE III. Sampling plan C.

Lot Size	Sample Size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.2.1 Door, rear (firing pin housing assembly) (see dwg. 9256082 covering a detail of dwg. 9256067 and table IV).

TABLE IV. Rear door, firing pin housing assembly.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Location of mounting hole	gage	III
102.	Thickness of base	gage	III
103.	Location of pull pin hole	gage	III
104.	Diameter of mounting hole	gage	III
105.	Protective finish incomplete or inadequate	visual	III

4.5.2.2 Tube (see dwg. 9256064 covering a detail of dwg. 9256067 and table V.

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TABLE V. Tube.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Inside diameter of tube	gage	III
102.	Outside diameter of tube	gage	III
103.	Outside diameter of tube shoulders	gage	III
104.	True position of shoulders	gage	III
105.	Protective finish incomplete or inadequate	visual	III

4.5.2.3 Inner tube assembly (see dwg. 9256067 and table VI).

TABLE VI. Inner tube assembly.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Straightness of tube	gage	II
102.	Location of center support	gage	II
103.	Parts loose	visual/ manual	II

4.5.2.4 Support, rear (see dwg. 9256065 covering a detail of dwg. 9256067 and table VII.

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TABLE VII. Rear support.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Outside diameter (OD)	gage	III
102.	Inside diameter (ID)	gage	III
103.	Location of screw holes	gage	III
104.	Depth to end of igniter tube slot	gage	III
105.	Depth of counterbore, minimum	gage	III
106.	True position of small ID to Datum A	gage	III
107.	Protective finish incomplete or inadequate	visual	III

4.5.2.5 Support, front (see dwg. 9256066 covering a detail of dwg. 9256067 and table VIII).

TABLE VIII. Front support.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Outside diameter	gage	II
102.	Inside diameter	gage	II
103.	True position of ID to Datum A	gage	II
104.	Protective finish incomplete or inadequate	visual	II

4.5.2.6 Support, center (see dwg. 9256068 covering a detail of dwg. 9256067 and table IX).

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TABLE IX. Center support.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	True position of small OD to Datum A	gage	III
102.	True position of large OD to Datum A	gage	III
103.	Small OD	gage	III
104.	Large OD	gage	III
105.	ID	gage	III

4.5.2.7 Screw, rear door (see dwg. 9256081 covering a detail of dwg. 9256067 and table X).

TABLE X. Rear door screw.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Pitch diameter of thread	gage	III
102.	Diameter of shoulder	gage	III
103.	Height of hole from base of head	gage	III
104.	Diameter of hole	gage	III
105.	Evidence of locking patch	visual	III
106.	Protective finish incomplete or inadequate	visual	III

4.5.2.8 Pin, rear door (see dwg. 9256083 covering a detail of dwg. 9256067 and table XI).

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TABLE XI. Rear door pin.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Diameter of pin	gage	II
102.	Height of pin (0.188-0.005)	gage	II
103.	Protective finish incomplete or inadequate	visual	II

4.5.2.9 Pin assembly, rear door (see dwg. 9256080 covering a detail of dwg. 9256067 and table XII).

TABLE XII. Pin assembly, rear door.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Length of lead assembly	gage	I

4.5.2.10 Label (M190) (see dwg. 9256085, covering a detail of dwg. 9256067 and table XIII).

TABLE XIII. Label.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Marking incorrect, incomplete or illegible	visual	I

4.5.2.11 Box, fiberboard, packing, ammunition for inner tube assembly and conversion kit (see dwg. 9256076 and table XIV).

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TABLE XIV. Box, fiberboard, packing.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Details missing from conversion kit bag	visual	II
102.	Conversion kit bag missing	visual	II
103.	Sealing strip loose, torn, incomplete, wrinkled or improperly adhered	visual	II
104.	Waterproof seal broken by puncture, cut or tear in moisture-proof bag	visual	II

4.5.2.12 Box, packing, ammunition for inner tube assembly and conversion kit (see dwg. 9256075 and table XV).

TABLE XV. Box, packing.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major:			
101.	Board broken or split	visual	I
102.	Binding strap missing or loose	visual	I

4.5.3 Testing.

4.5.3.1 First article sample (see 4.4). The samples submitted as the first article quantity shall be tested as specified herein after being examined 100 percent for the applicable classifications of defects in paragraph 4.5.2.

4.5.3.1.1 Pull test of inner tube assembly (see 3.4 and dwg. 9256067). A sample of 10 inner tube assemblies shall be subjected to the pull test specified in 4.6.1 using test equipment specified in 4.5.4. Rejection shall be as specified in 4.4.3.

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4.5.3.2 Production.

4.5.3.2.1 Pull test of inner tube assembly (see 3.4 and dwg. 9256067). A sample of 10 inner tube assemblies shall be subjected to the pull test specified in 4.6.1 using test equipment specified 4.5.4. The lot shall be rejected if one or more inner tube assemblies are classed defective.

4.5.4 Inspection equipment. Equipment Tabulation ET-9256067 identifies the inspection equipment required to perform the examinations and tests prescribed in this section.

4.5.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.5.4.2 Calibration. The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with MIL-STD-45662.

4.6 Test method and procedures. The test methods and procedures described in each of the following tests represents only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using substitute methods and procedures only after obtaining Government approval. Government approval will be based on an evaluation to determine that the substitute method and procedure will result in an equal or better quality evaluation than that specified herein.

4.6.1 Pull test of inner tube assembly. The front and rear supports of the inner tube assembly shall withstand an axial pulloff force of the specified pounds after swaging of the inner tube (see 3.4).

4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of drawings 9256076 and 9256075 and MIL-STD-1190.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking will be Level C in accordance with MIL-STD-1190. Inner tube assemblies and installation kits are to be packed in accordance with drawings 9256075 and 9256076.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification addresses special requirements (not normally provided on drawings) and quality assurance provisions for the fabrication of parts, assembly and packing of an inner tube assembly and conversion kit used with the M73 35mm practice rocket.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Data cards shall be prepared for each lot in accordance with specified Data Item Descriptions and MIL-STD-1167

d. Instructions for the submission of first article samples (see 3.3)

e. Instructions for the delivery of test data hardware and inspection records.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Inspection and test data. Two copies of the inspection and test data for each lot are to be forwarded to the Government in accordance with the appropriate contract.

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Launcher, rocket
Packaging, ammunition
Rocket, practice, M73

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-50857A (MI)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
INNER TUBE ASSEMBLY AND CONVERSION KIT, PARTS FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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